



Mitigation of vortex-induced vibrations in bridges under conflicting objectives

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ABSTRACT

Aerodynamic performance of long-span bridges can be improved by retrofitting cross-sectional shapes or by adding tuned mass dampers. When both the approaches are used simultaneously, the design problem may become too complicated to solve by engineering judgment. The complexity becomes even greater in a real world problem in which both the performance and the cost have to be considered. We propose a multi-objective based approach to obtain optimal solution of the design problem. The approach effectively deals with highly nonlinear design space and the conflicting objectives between the performance and the cost. Due to the conflict of the two objectives, multiple optimal solutions are obtained, which are called Pareto-optimal solutions. The final design is obtained considering the cost and the performance criteria. The proposed method is applied to the mitigation of vortex-induced vibrations, but its general framework is applicable to any other aerodynamic problems.

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1. Introduction

Wind loading is one of the important factors that needs due consideration in designing long span bridges. Over the past few decades, the maximum span lengths have increased significantly. The Akashi-Kaikyo Bridge has the central suspended span of 1991 m. Even longer bridges, such as Messina Strait Bridge with 3300 m span length and Gibraltar Strait Bridge with 5000 m span length, are planned (Jones et al., 2003). The increased span length has attracted one drawback in terms of the slenderness of bridge decks, which is more susceptible to wind-induced vibrations.

Wind effects are rigorously investigated in designing long span bridges. Most bridges do not experience serious wind-related problems. However, some bridges have experienced significant wind-related problems after the construction, which were not anticipated in the design stage. One solution to attenuate the aerodynamic response of an existing bridge is to retrofit the cross section and thereby improving aerodynamic response. Larsen et al. (2000) have shown that adding guide vanes to the main span of girder decreased the vortex-induced oscillations in the Greatbelt Suspension Bridge. The Bronx-Whitestone Bridge (Barrelli et al., 2006) and Lions' Gate Bridge (Vejrum et al., 2000) also retrofitted

sections in part to improve the aerodynamic performance. In situations where structural modifications prove ineffective or impractical, dampers are installed to reduce the vibrations. Examples of bridges that used tuned mass dampers (TMD) to mitigate the vortex-induced vibrations are the Rio-Niteroi Bridge (Battista and Pfeil, 2000) and the Trans-Tokyo Bay Crossing Bridge (Fujino and Yoshida, 2002).

This paper investigates a problem in which both cross-sectional retrofit and tuned mass dampers are used to mitigate the vortex-induced vibrations. The design problem also considers both the performance and the cost of the retrofit, thereby increasing the complexity of the problem. The design problem typically is too complex to be solved by engineering judgment.

We formulate the design problem as an optimization problem. The objectives of the optimization are to improve the performance and to reduce the cost. To deal with the conflicting objectives between the performance and the cost, we employ a multi-objective optimization technique. The multi-objective optimization provides multiple candidate solutions, which are called Pareto-optimal solutions. Once the Pareto-optimal solutions are obtained, a decision can be made by comparing the cost and the performance in a systematic way.

This paper first presents a brief overview of vortex shedding analysis of a bridge deck along with the analysis of cross-sectional retrofit and tuned mass dampers. In the next section, conflicting objectives in the retrofit problem are explained. After reviewing

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the multi-objective optimization technique employed in this paper, the proposed method is discussed with the help of an illustrative example.

2. Analysis of mitigation options for vortex-induced vibrations

In order to reduce the vortex-induced vibrations, cross-sectional retrofit, or tuned mass dampers are commonly used. In this paper, frequency-domain analysis is employed to analyze the vortex-induced vibrations. The effect of cross-sectional retrofit is considered by changing the aerodynamic coefficients. The effect of tuned mass dampers is considered using the effective damping.

2.1. Analysis of vortex-induced vibrations

Vortex-induced vibration is an aerodynamic response that may occur to a flexible structure subjected to cross wind. The separation of air flow causes vortices to shed alternatively on either side of the bridge deck, which in turn causes vibrations. Experimental evidence shows that these fluctuating loads are narrow banded and centered at vortex shedding frequency. The vortex shedding frequency is expressed as

$$f_s = S_t \frac{V}{D} \quad (1)$$

in which S_t is the Strouhal number, V is cross wind velocity, and D is cross-sectional depth. The vibration frequency does not change even when the wind speed increases slightly, which is called “lock-in” state as shown in Fig. 1. When the wind speed increases from V_1 to V_2 , the vibration frequency stays at f_1 .

For the analysis of vortex-induced vibrations, a frequency-domain formulation (Strømmen, 2006) is employed. The equation of motion in the frequency domain is

$$(-\tilde{M}_i \omega^2 + \tilde{C}_i i\omega + \tilde{K}_i) \cdot a_{\eta_i}(\omega) = a_{\tilde{Q}_i}(\omega) + a_{\tilde{Q}_i^{ae}}(\omega, \eta_i, \dot{\eta}_i, \ddot{\eta}_i) \quad (2)$$

in which a_{η_i} , $a_{\tilde{Q}_i}$, and $a_{\tilde{Q}_i^{ae}}$ are Fourier amplitudes of the modal coordinates $\eta_i(t)$, the modal flow induced load $\tilde{Q}_i(t)$, and the modal motion induced load (due to the interaction between flow and structural motion) $\tilde{Q}_i^{ae}(t)$, respectively. By assuming that the motion induced load is proportional to and in phase with the structural response, and by arranging the equations accordingly,

$$a_{\eta_i}(\omega) = \frac{\hat{H}_i(\omega)}{\tilde{K}_i} a_{\tilde{Q}_i}(\omega) \quad (3)$$

in which $\hat{H}_i(\omega)$ is the modal frequency-response function. The response spectrum for the displacement at $x=x_r$ is given by

$$S_{r_i}(x_r, \omega) = \phi_i^2(x_r) \frac{|\hat{H}_i(\omega)|^2}{\tilde{K}_i^2} S_{\tilde{Q}_i}(\omega) \quad (4)$$

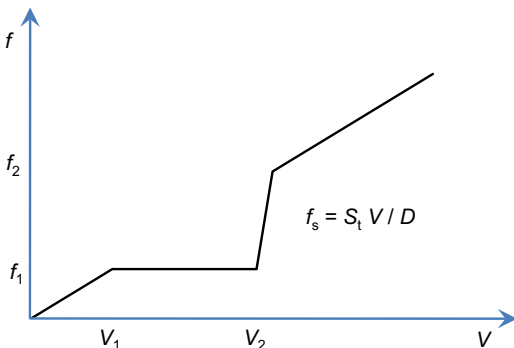


Fig. 1. Response characteristics under vortex shedding.

in which $\phi_i(x)$ is the mode shape. The modal frequency-response function $\hat{H}_i(\omega)$ includes the interaction between flow and structural motion, and therefore, includes the aerodynamic derivatives H_k^*, A_k^* ($k=1 \sim 6$). The load spectra $S_{\tilde{Q}_i}(\omega)$ represent the flow induced load. The spectra proposed by Vickery and Basu (1983a, 1983b), following the notation in Strømmen (2006), are employed. The vertical and rotational load spectra and the corresponding co-spectra are as given in Eqs. (5) and (6), respectively.

$$\begin{bmatrix} S_{q_z}(\omega) \\ S_{q_\theta}(\omega) \end{bmatrix} = \frac{\left(\frac{1}{2} \rho V^2\right)^2}{\sqrt{\pi} \omega_s} \begin{bmatrix} \frac{(B \hat{\sigma}_{q_z})^2}{b_z^2} \exp\left\{-\left(\frac{1-\omega/\omega_s}{b_z}\right)^2\right\} \\ \frac{(B^2 \hat{\sigma}_{q_\theta})^2}{b_\theta^2} \exp\left\{-\left(\frac{1-\omega/\omega_s}{b_\theta}\right)^2\right\} \end{bmatrix} \quad (5)$$

$$\hat{C}o_{q_m}(\Delta x) = \cos\left(\frac{2\Delta x}{3\lambda_m D}\right) \exp\left\{-\left(\frac{\Delta x}{3\lambda_m D}\right)^2\right\} \quad (6)$$

in which ρ is air density, $\omega_s = 2\pi f_s$, B is the width of the cross section, D is the depth of the cross section, $\hat{\sigma}_{q_m}$ is the non-dimensional root mean square lift or torsion moment coefficient, b_m is the non-dimensional load spectrum band width parameter, λ_m is the non-dimensional coherence length scale, and Δx is the span-wise separation.

Once the response spectrum is obtained, the variance of the displacement is calculated by integrating the response spectrum. The maximum displacement is obtained by multiplying the peak factor by the standard deviation of the displacement. Therefore, the maximum displacement at $x=x_r$ for certain mode is given by

$$r_i(x_r) = k_p \left(\int_0^\infty S_{r_i}(x_r, \omega) d\omega \right)^{1/2} \quad (7)$$

in which the peak factor k_p for a narrow banded process such as vortex-induced vibrations is $\sqrt{2}$. Further details of the derivations and relevant equations can be found from (Strømmen, 2006). In order to combine the modal contributions, the square root of sum of squares (SRSS) rule is used.

$$r_0(x_r) \approx \left(\sum_i \{r_i(x_r)\}^2 \right)^{1/2} \quad (8)$$

2.2. Mitigation of vibrations by the cross-sectional retrofit

The aerodynamic response of a bridge deck can be controlled using aerodynamically favorable cross-section. When air flows around the aerodynamically favorable section, the flow separation is minimized. In general, the cross-section determined during the design stage remains throughout the life of a bridge. However, retrofitted cross-sections were employed for some bridges after the construction, to deal with aerodynamic problems that were not anticipated during the design stage. Examples of such bridges were provided in the introduction.

In order to analyze the effect of the cross-sectional retrofit, aerodynamic derivatives and load spectra shown in Eqs. (5) and (6) are needed for the new cross-section. Aerodynamic derivatives of the new cross-section represent change of the characteristics of the interaction between flow and structural motion. Specifically, new aerodynamic derivatives H_k^*, A_k^* ($k=1 \sim 6$) are used for the frequency response function $\hat{H}_i(\omega)$ of Eq. (4). Change in the cross-sectional shape significantly changes the aerodynamic derivatives. For example, see Matsumoto et al. (1996), Noda et al. (2003), and Simiu and Scanlan (1996). For the new cross-section, load spectra also should be updated. In Eqs. (5) and (6), $\hat{\sigma}_{q_m}$, b_m , λ_m , and S_t of the new section are used. Recommended values of the first three coefficients are provided in Strømmen (2006). The

Strouhal number changes significantly due to the change in shape or angle. For examples, see Norberg (1993), Chen and Fang (1996), Matsumoto (1999), and Simiu and Scanlan (1996).

2.3. Mitigation of vibrations by tuned mass dampers

In addition to the cross-sectional retrofit, tuned mass dampers may also be used to mitigate the vibrations. In this paper, we assumed that the both the cross-sectional retrofit and the tuned mass dampers are effective in mitigating the vibrations. However, we note that exceptions exist such as the Trans-Tokyo Bridge in which tuned mass dampers were found to be more effective than the cross-sectional retrofit during the wind tunnel test (Fujino and Yoshida, 2002).

The control effects of the tuned mass dampers have been investigated from interaction analysis of the system including bridges, tuned mass dampers, and external motion dependent loads (Kwon and Park, 2004), or from simply adopting the additional effective damping introduced by installation of tuned mass dampers. Due to the complicated procedure of the vortex-induced vibrations, tuned mass dampers are modeled using the effective damping. After manipulating the equations for two degree of freedom system under harmonic force excitation (Warburton, 1982), the effective damping can be given as (Yamaguchi, 1996)

$$\zeta_{eff} = \frac{1}{2} \sqrt{\frac{\mu/2}{1+\mu/2}} \quad (9)$$

$$\mu = \frac{m_t}{m_i} \quad (10)$$

in which m_t is the mass of the damper and m_i is the modal mass. Once the effective damping is obtained, both the structural damping ζ_s and the effective damping ζ_{eff} are used for the frequency response function $\hat{H}_i(\omega)$ of Eq. (4), i.e., $\zeta_i = \zeta_s + \zeta_{eff}$ is used for each mode. Note that $\hat{H}_i(\omega)$ also includes the aerodynamic damping ζ_i^{ae} , which is subtracted from ζ_i to consider the interaction of the flow and the structural motion.

3. Objectives of the mitigation: minimization of the vibration and the cost

The primary objective of the mitigation is to reduce the wind-induced vibrations. Formally stated, the objective function is

$$f_1(\mathbf{X}) = r_0^{\max} \quad (11)$$

in which $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_{N_r}, d_1, d_2, \dots, d_{N_d}\}$, $\mathbf{x}_k (k=1, \dots, N_r)$ is the bound (coordinates of beginning and end) of the k th cross-sectional retrofit, N_r is the number of bounds to retrofit cross-sections, $d_k (k=1, \dots, N_d)$ is the number of dampers at the k th location, N_d is the number of locations to install dampers, and $r_0^{\max}$ is the maximum displacement of the bridge obtained from Eq. (8). The goal of the mitigation is to find $\mathbf{X}$ that minimizes $f_1(\mathbf{X})$.

In a real world problem, the cost of the retrofit becomes the second objective of the design, further complicating the problem. The total cost of the retrofit is the cost of cross sectional retrofit plus the cost of dampers. Therefore, the second objective function that also should be minimized is

$$f_2(\mathbf{X}) = \sum_k C_r \text{length}(\mathbf{x}_k) + \sum_k C_d d_k \quad (12)$$

in which C_r is the cost of retrofit per unit length, and C_d is the cost of one damper.

The objectives to be achieved here are reduction in response with minimum cost. If the displacement is required to be reduced,

the corresponding cost will increase and vice versa. Thus, the approach to mitigate the aerodynamic response results in conflicting objectives. The two objectives conflict with each other and retrofit cannot be achieved based on only one of the objectives. The design problem has high dimension of $\mathbf{X}$ and two conflicting objectives, which makes it difficult to be solved by engineering judgment.

4. The proposed approach

We formulate the design problem as an optimization problem. The objectives of the optimization are to minimize the objective functions given in Eqs. (11) and (12). Design variables are given in $\mathbf{X}$. If we solve the optimization problem, the optimal $\mathbf{X}$ will be found that minimizes both the objective functions.

A simple approach to deal with the multiple objective functions is to use a weighted single objective function. Minimization is done for a scalar function, i.e.,

$$\min F(\mathbf{X}) = \min \sum_i w_i f_i(\mathbf{X}) \quad (13)$$

in which w_i are weight factors, which determine the relative importance of different objective functions. In a practical situation it may not always be feasible to determine the relative importance of one objective over the other. Also, arbitrary weights may not always lead to the true optimal solution. The other way to address the problem using single objective formulation is to run the optimization for various combinations of w_i , but this approach is computationally too expensive.

We employ a multi-objective formulation to simultaneously minimize the objective functions, i.e.,

$$\min F(\mathbf{X}) = \min [f_1(\mathbf{X}) \quad f_2(\mathbf{X}) \quad \dots \quad f_{n_{obj}}(\mathbf{X})] \quad (14)$$

The design problem solved in this paper has $n_{obj}=2$. The overall process in solving this multi-objective optimization problem is shown in Fig. 2. Randomly generated design trials $\mathbf{X}_i (i=1, \dots, n)$ will be gradually improved during the optimization. The optimization requires the evaluation of the objective functions $f_1(\mathbf{X})$ and $f_2(\mathbf{X})$ (Eqs. (11) and (12)), details of which were explained in Sections 2 and 3. The rest of the steps involve the multi-objective genetic algorithm, which is explained in the next section.

5. Overview of multi-objective optimization

The multi-objective optimization shown in Eq. (14) minimizes multiple objective functions simultaneously. Due to the conflicting nature of the objectives, no single solution can be termed as the best solution. As an illustration, see Fig. 3 which shows optimal solutions from multi-objective optimization. Each point corresponds to a design variable X in Eqs. (11) and (12). The solution A is better than the solution B in terms of the deflection, but the solution B is better than the solution A in terms of the cost. Therefore, neither A nor B can be selected as a single optimal solution, but both of them should be considered together as part of optimal solutions. Such optimal solutions are called as Pareto-optimal solutions or Pareto-optimal front.

On the other hand, the solution D is clearly poorer than the solution B in both the objectives, so it cannot be said as an optimal solution. The multi-objective optimization is not fully converged when it can still improve the existing candidates. Since multiple solutions should be compared simultaneously during the convergence, algorithms that minimize a group of solutions rather than a single solution are ideal for multi-objective

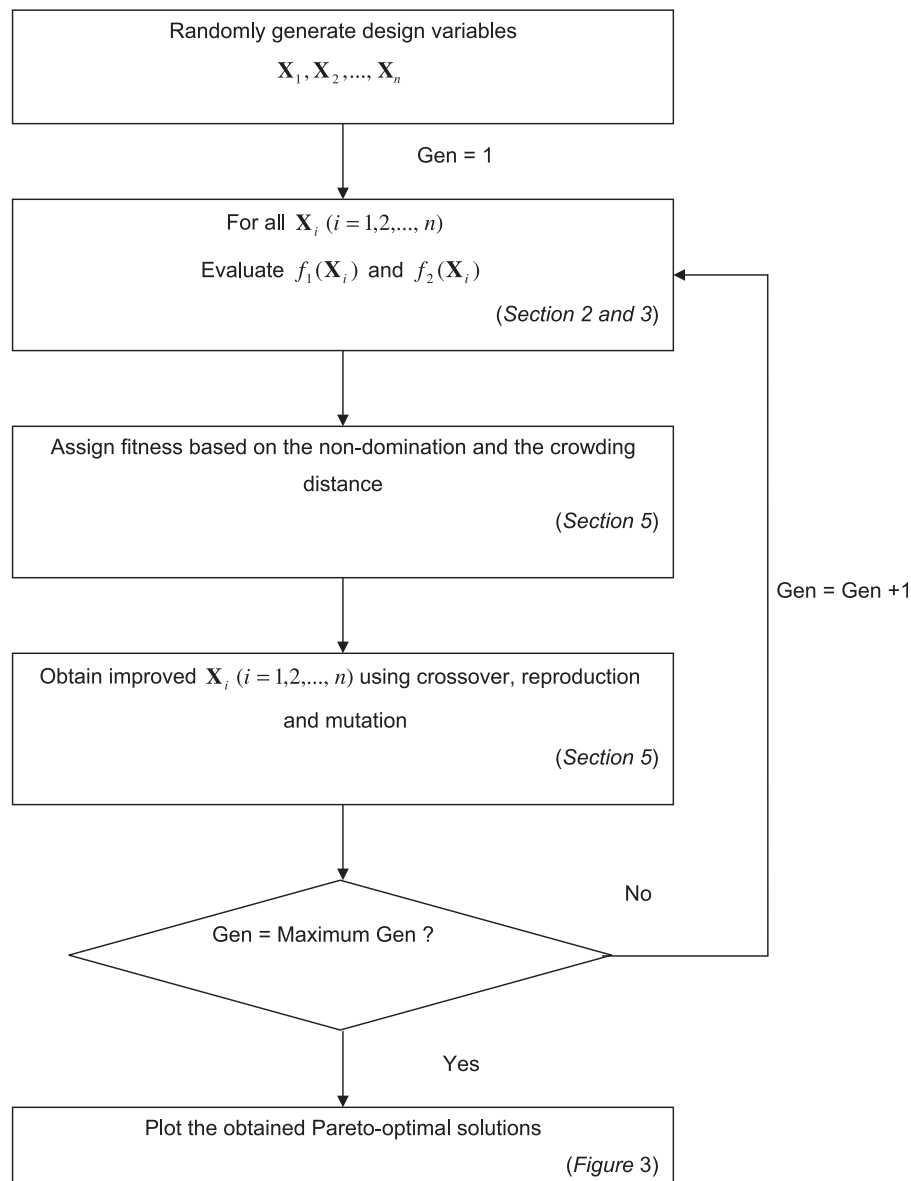


Fig. 2. Flow chart of the proposed method.

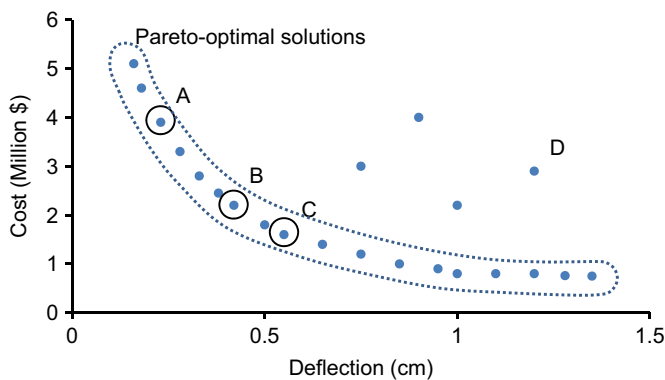


Fig. 3. Schematic illustration of Pareto-optimal solutions in multi-objective optimization.

optimization. The genetic algorithm is such an algorithm (Goldberg, 1989). The genetic algorithm was inspired by natural evolution where the offspring inherits qualities from the parents. Similar to the evolution, multiple individuals are optimized

simultaneously to better “fit” the given objective functions. A better-fit individual has a better chance to transfer its characteristics to the offspring. A generation is the computational step whereby new set of individuals are produced based on the fitness of existing individuals. Many generations are usually necessary for the convergence. The genetic algorithm uses three main operators—reproduction, crossover, and mutation. Further details of the genetic algorithm can be found from Goldberg (1989).

In this paper, we have employed the non-dominated sorting genetic algorithm (NSGA-II) by Deb et al. (2002), which uses the concept of the non-domination to find the optimal solutions. Fig. 4 illustrates the non-domination. The goal of the optimization is to minimize both the objective functions f_1 and f_2 . A solution S_2 is termed as dominated by another solution S_1 , if S_2 is not better than S_1 in all the objectives but is inferior to S_1 in at least one objective. In the figure, S_2 and S_4 are dominated by S_1 , S_3 , and S_5 . The solutions that are not marked as dominated are termed as non-dominated solutions. All these solutions constitute the first non-dominated front. The NSGA-II also creates the second level of front, the third, etc, till all the candidates in the solution space are exhausted. After marking the individuals according to the non-domination level, the

fitness is assigned to each individual accordingly. Along with the non-domination, the NSGA-II also uses the crowding distance to improve the diversity. The crowding distance operator compares two solutions to find the better solution. A solution with better rank and larger crowding distance wins over the other solutions. A solution with larger crowding distance resides in a less crowded part of the solution space and thus imparts diversity in contesting solutions. For further details, see Deb et al. (2002) and Deb (2003, 2008).

In theory, the multi-objective optimization can deal with any number of objective functions. In practice, however, it is not practical to deal with more than three objective functions. Ongoing efforts such as Brockhoff and Zitzler (2006) and Deb and Saxena (2006) may enable us to optimize for any number of objective functions in the near future. In this paper, only two objective functions were used and the convergence was not an issue.

6. An illustrative example

6.1. Description of the problem

In order to solve a realistic design problem, the bridge for illustration is modeled after the Rio-Niterói Bridge (Battista and Pfeil, 2000). Fig. 5 shows schematic drawings of the bridge and the cross-section. The total length of the bridge is 850 m, with 3 long spans and 2 short spans. The width and the depth of the cross-section are 25.9 m and 7.42 m, respectively. The two box girders of the original Rio-Niterói Bridge are simplified in creating the cross-section. The original bridge experienced vortex-induced vibrations

in the vertical mode at about 15 m/s. It installed tuned mass dampers to mitigate the vibrations. In this paper, we assume that the vibrations of the bridge are mitigated using both the cross-sectional retrofit and the tuned mass dampers. The retrofitted cross-section is also shown in Fig. 5, in which double flaps are used.

For the analysis, three vertical modes are considered. Fig. 6 shows the three normalized modes for the hypothetical bridge. The frequencies and the modal mass corresponding to these three modes are given in Table 1. The coefficients in Eqs. (5) and (6) are assumed to be the values given in Table 2.

6.2. Wind tunnel test

The experimental set up for the wind tunnel test is shown in Fig. 7. The simplified section model of the Rio-Niterói Bridge corresponded to 63 m long girder of the full bridge at a prototype

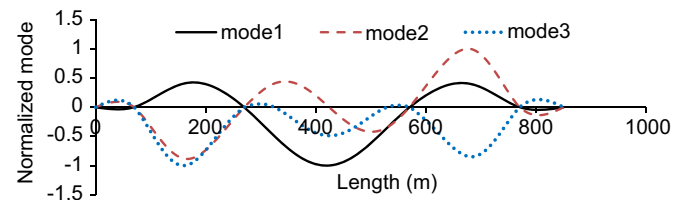


Fig. 6. Mode shapes of the bridge.

Table 1

Mode data for the vortex analysis of the bridge.

Mode no	Frequency (Hz)	Modal mass (ton)
1	0.315	2865
2	0.624	2553
3	0.768	2189

Table 2

Coefficients used for the analysis of vortex-induced vibrations.

Coefficient	Original section	Retrofitted section
S_t	0.145	0.1
S_{gz}	0.4	0.1
b_z	0.15	0.1
λ	2	2

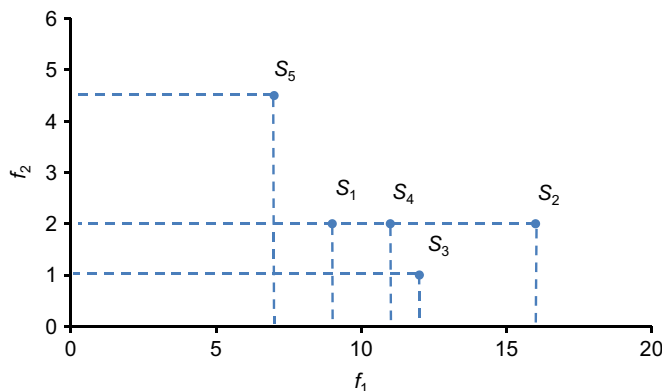


Fig. 4. Non-dominance used in the NSGA-II.

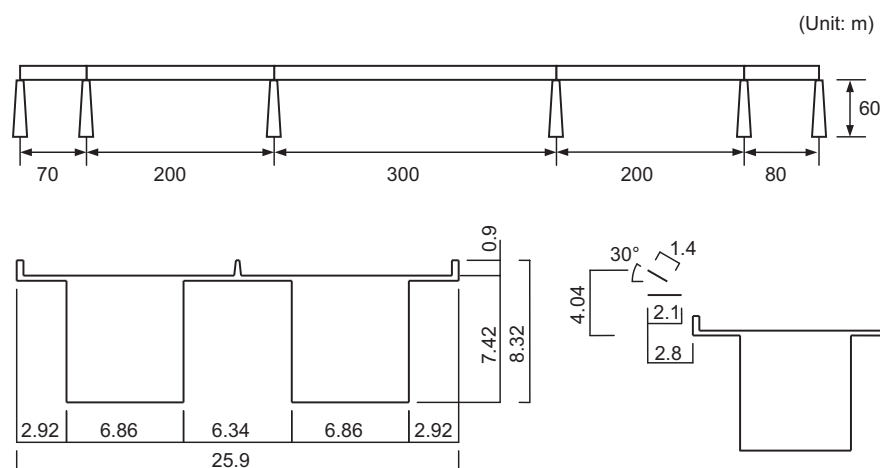


Fig. 5. Schematic drawings of the bridge and the cross-section (modeled after the Rio-Niterói Bridge (Battista and Pfeil, 2000)). Left: original cross-section, right: retrofitted cross-section with double flaps.

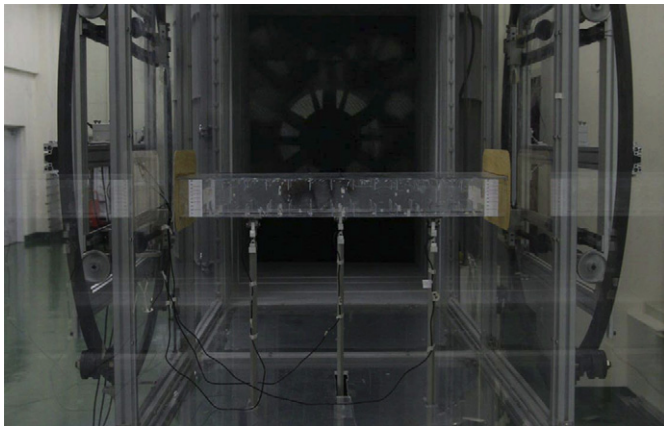


Fig. 7. Experimental set up for wind tunnel testing of bridge deck.

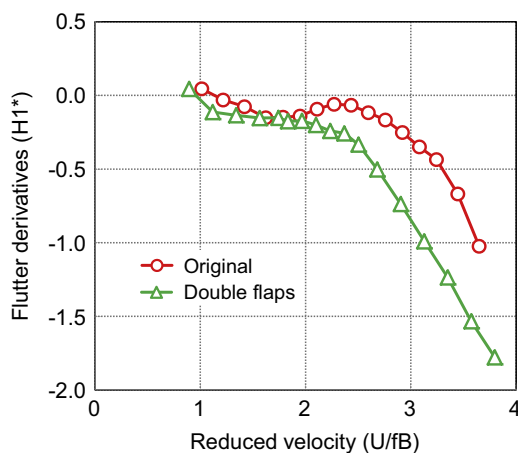


Fig. 8. Aerodynamic derivatives (H_1^*) obtained from wind tunnel tests.

scale of 1–70. The flutter derivatives which describe the motion dependent aerodynamic forces were measured from the forced vibration technique (Lee and Kwon, 2009). It is generally accepted that the amplitude of flow-induced vibrations would decrease with increasing turbulence intensity, and the tests were therefore conducted in the smooth flow to clarify the vortex-induced vibration. The aerodynamic derivatives related to the vertical motion (H_1^*) are as shown in Fig. 8. The line with circles indicates the aerodynamic derivative for the original section whereas the line with triangles indicates the aerodynamic derivative for the retrofitted section. There is a hump at the H_1^* curve for the original section around reduced wind velocity of 2.4 where the vertical vortex-induced vibration occurs. The retrofitted section used double flaps, which effectively removed the vortex-induced vibration.

6.3. Baseline analysis

Before conducting the optimization, the bridge without the retrofit is analyzed to understand the behavior. The result is shown in Fig. 9. The response is calculated at the various locations “D”, which are measured from the left-end of the bridge. For example, $D=425$ m corresponds to the center of the entire bridge. The three locations plotted in the figure correspond to the centers of the 3 long spans. Vortex-induced vibrations occur at about 17 m/s, which is slightly higher than 15 m/s of the original Rio-Niterói Bridge. Additional vibrations occur at higher speeds.

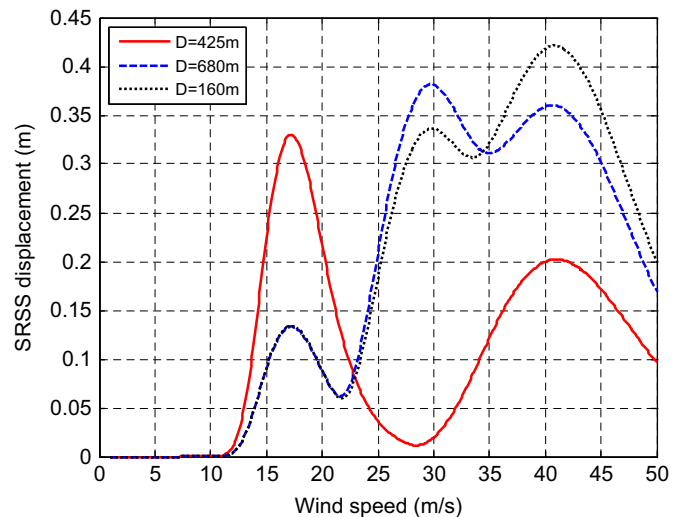


Fig. 9. Aerodynamic response before retrofit.

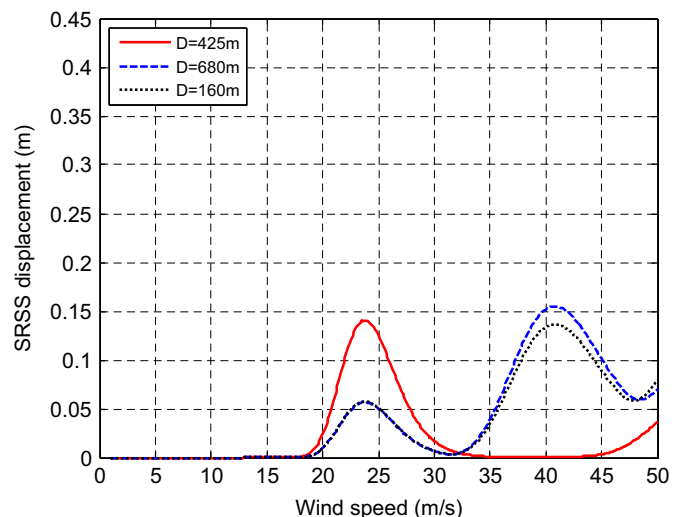


Fig. 10. Aerodynamic response when all cross sections are retrofitted.

To understand the effect of the cross-sectional retrofit, another model is analyzed with aerodynamic coefficients and parameters of the retrofitted section. The entire bridge section is assumed to be retrofitted. The result is shown in Fig. 10. Similarly, the same bridge is analyzed again with only TMDs without the cross sectional retrofit. All the three sections were modeled with two TMDs at the center of each section. The result is shown in Fig. 11.

It is evident that the reduction in displacement is satisfactory if either the entire cross sections are retrofitted or 6 TMDs are added, but these solutions would involve high cost. In order to find the solution that has satisfactory performance and cost, many trials like these are necessary especially when we use both the cross-sectional retrofit and the tuned mass dampers. The proposed multi-objective optimization approach is applied to this design problem.

6.4. Optimization and Pareto-optimal solutions

In order to identify cost-effective retrofit, multi-objective optimization is performed with two objective functions shown

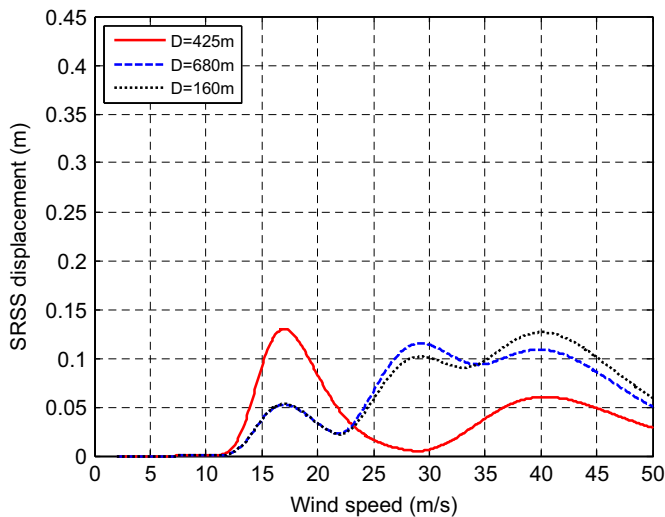


Fig. 11. Aerodynamic response with TMDs only.

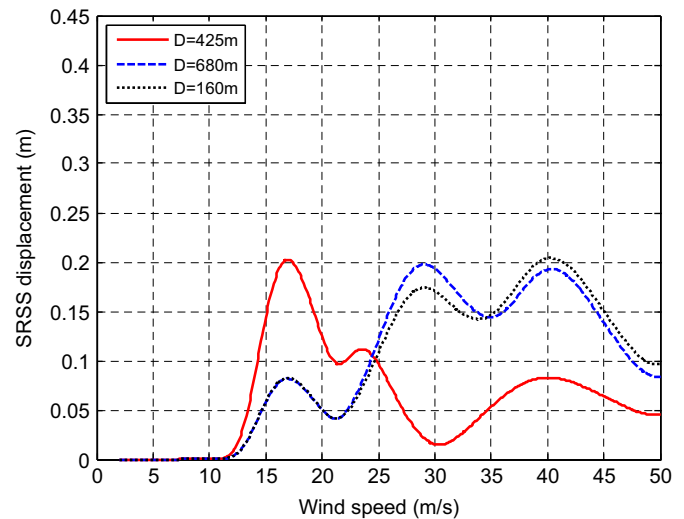


Fig. 13. Aerodynamic response of the optimized solution B of Fig. 12.

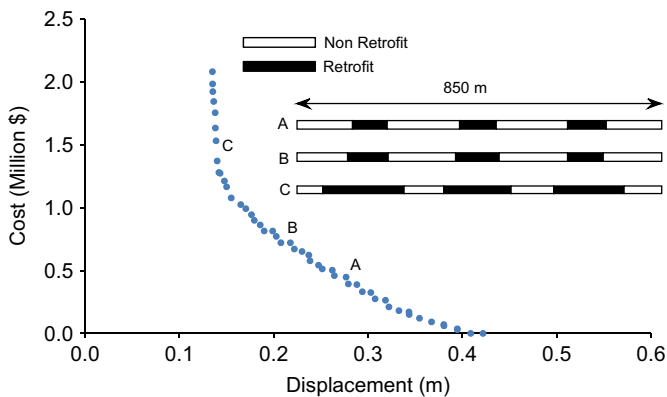


Fig. 12. Optimized solution—cross sectional retrofit only.

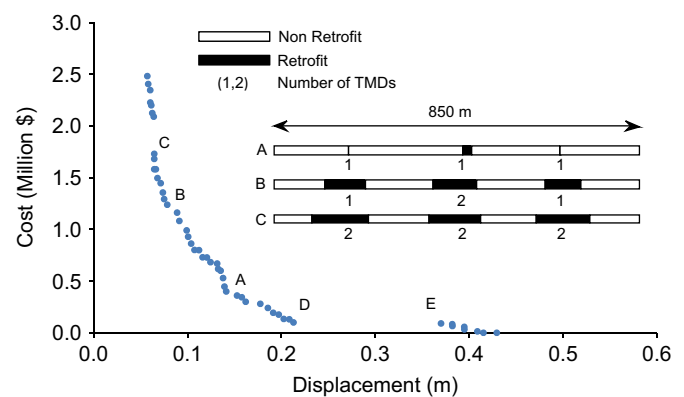


Fig. 14. Optimized solution—cross sectional retrofit and TMDs.

in Eqs. (11) and (12). In the first optimization analysis, only the cross-sectional retrofit is considered, i.e., $N_r=3$ and $N_d=0$ are used. The domains of $x_k(k=1,\dots,N_r)$ are set such that they correspond to the three long spans. After the optimization, their optimum bounds will be obtained. The NSGA-II algorithm is used with 50 individuals' 2000 generations. The cost of the retrofit is assumed as $C_r=3000$ dollars per meter and $C_d=100,000$ dollars per unit.

The Pareto-optimal solutions obtained from the optimization without TMDs are shown in Fig. 12. The Pareto-optimal solutions quantify the relationship between the cost and the performance, thereby enabling performance-based decision. A user can determine the best solution that meets the performance and the cost need of the user. Each solution of the Pareto-optimal solutions provides an optimal design. Further details of the three solutions A, B, and C are explained to illustrate the use. The three bars within the figure schematically show the bounds where the retrofit is conducted. For the solution point A, the cost of the retrofit is \$0.45 million, and the displacement is 0.277 m. For the solution point B, the cost is increased to \$0.77 million, but the displacement is further reduced to 0.203 m. For the solution point C, the displacement is reduced to 0.139 m, but the cost of the retrofit is as high as \$1.53 million. The user can arrive at a

particular solution based on the cost and the performance. For any solution, a response plot can be drawn to assist the decision-making. An example response plot is shown in Fig. 13 for the solution B, which plots the displacement for various wind speeds.

In addition to the cross-sectional retrofit, TMDs can also be used to reduce the displacement. Addition of TMDs does not change the number of objective functions, but increases the design space shown in Eq. (11). In addition to the design variables for cross-sectional retrofit, the number of TMDs is also a design variable, i.e., $N_r=3$ and $N_d=3$ are used. The optimization problem becomes more complicated due to the increased design space.

Fig. 14 shows the result when the TMDs are used along with the cross-sectional retrofit. All the points along C, B, A, D, E curve are the optimized solutions under the two conflicting objectives. Example solutions A, B, and C show the number of TMDs in addition to the cross-sectional retrofit. Point C indicates small displacement with high cost and point A indicates low cost with comparatively larger displacement. The sudden gap between the points D and E is due to the discontinuity in the design space. Unlike the cross-sectional retrofit, the number of TMDs is a discrete variable, resulting in the discontinuity in the Pareto-optimal solutions. The design D uses 1 TMD whereas the design E does not use any TMD.

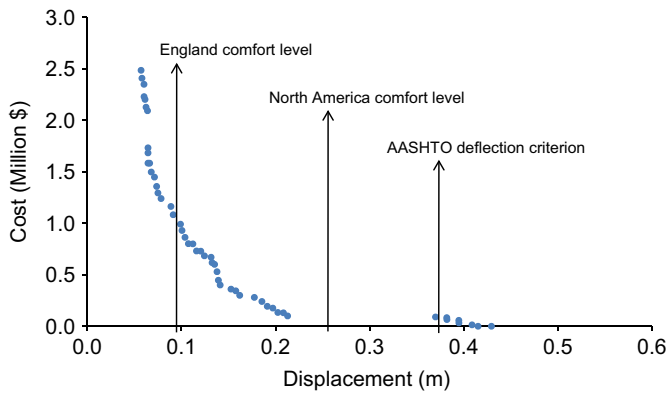


Fig. 15. Comparison with acceptance criteria for deflection and acceleration.

6.5. Acceptance criteria for the allowable displacement

After obtaining the Pareto-optimal solutions, the next step is to select the solution based performance requirement and the cost. The acceptance criterion of the displacement is very helpful in the decision-making. This section discusses some of the available criteria for the allowable displacement.

For the bridges in the United States, the deflection criterion is given in ASSHTO 4th Edition (2007). As per the clause C2.5.2.6.2, the deflection limit for vehicular and general purpose bridge is span/800. For vehicular and pedestrian load the deflection limit is restricted to span/1000.

Another criterion to consider is the acceleration criterion related to the passenger comfort. For example, Gu et al. (2002) have proposed formula for the acceptable acceleration.

$$A_{C_{peak}} = 0.1g \exp(0.9163(V - V_d)/(V_d - 20)) \quad (15)$$

in which V = wind speed and V_d = designed wind speed. In North America (Irwin, 1999) the maximum peak acceleration should be less than 0.5 g for the wind speed less than 13 m/s, and less than 0.1 g for the wind speed between 13 m/s and the design wind speed of the bridge.

$$A_{C_{peak}} < 0.05g \quad \text{for } V < 13 \text{ m/s} \quad (16)$$

$$A_{C_{peak}} < 0.1g \quad \text{for } 13 \text{ m/s} < V < V_d \quad (17)$$

In England (Gu et al., 2002), the maximum peak acceleration should be less than 0.04 g for the wind speed lower than 20 m/s.

$$A_{C_{peak}} < 0.04g \quad \text{for } V < 20 \text{ m/s} \quad (18)$$

For steady state harmonic response, the acceleration criterion can be easily converted to the displacement criterion since the magnitude of the acceleration $\ddot{u}_0 = \omega^2 u_0$. It is reasonable to assume that the response due to the vortex-induced vibrations is harmonic. For the previous illustrative example, the natural frequency of the first mode $f = 0.315$ Hz. The displacement according to the North American Standard is computed as 0.26 m and that for the England Standard is 0.10 m.

Fig. 15 shows the optimized solution along with the acceptance criteria. As there are uncertainties in aerodynamic coefficients, other assumed parameters, or even the numerical analysis, these criteria should be used with flexibility. A design that barely satisfies the criteria may not perform exactly as expected in the field. If certain criteria should be satisfied strictly, conservative designs should be selected. The existence of uncertainties makes it very useful to use the multi-objective analysis to make the decision. Rather than selecting a single design that may not

satisfy the criteria or may be too conservative, the decision-making based on the proposed approach would result in a well-performing and economical solution.

6.6. Criteria for fatigue damage

Vortex-induced vibrations occurring for the life span of the bridge may cause fatigue-related problems (Hosomi et al., 1997, Repetto and Solari, 2002, Holmes, 2002). In this section, simplified fatigue analysis is conducted using the following steps. The analysis requires many equations, including probability integrals, and the reader is referred to a paper written by the second and the fourth author (Jung et al. 2011) for further details.

- (1) For certain wind speed, obtain the standard deviation of the displacement under vortex-induced vibrations using the approach explained in Section 2.1. Using the relations between the second derivative of the displacement and the stress, obtain the standard deviation of the stress.
- (2) Assuming that the peaks of the stresses follow the Rayleigh distribution (Strømmen, 2006, p.29), obtain the probability function of stress.
- (3) Obtain the probability of the wind for this wind speed.
- (4) Repeat (1)–(3) for multiple wind speeds, and obtain the probability for certain level of stress.
- (5) Repeat (1)–(4), and obtain the probability for all levels of stresses.
- (6) For all levels of stresses, compute the number of cycles = (probability from (5)) $\times$ (life of the bridge) $\times$ (frequency of vibration) $\times$ (probability of having the direction that causes the vortex shedding).
- (7) Use the Miner's rule to estimate the fatigue damage. If the fatigue damage is greater than 1, fatigue problems may arise.

Fatigue damage is computed at the center of the mid-span, for the top and the bottom of the box girders where the magnitude of the stress is the highest. Elastic modulus is assumed to be 200 GPa. The Rayleigh distribution is used, with mean wind speed at the deck height = 6.88 m/s (Kwon et al., 2011), to obtain the probability of having certain wind speed. Wind speeds that may cause the vortex shedding are included in the calculation ($V = 15 \sim 19$ m/s). Wind direction data was not available, so it was assumed that 5% of all directions fall into the angle that may cause vortex-induced vibrations. Fatigue was checked for AASHTO "Detail A" (AASHTO, 2007).

Fig. 16 shows the optimized solutions with fatigue life of 100 years, 50 years, and 30 years. When this figure is compared with Fig. 15, it is clear that the solutions that meet the North America

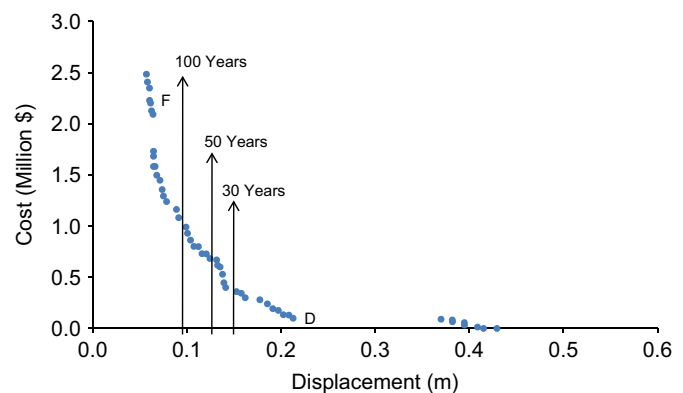


Fig. 16. Comparison with criteria for fatigue for AASHTO Detail A.

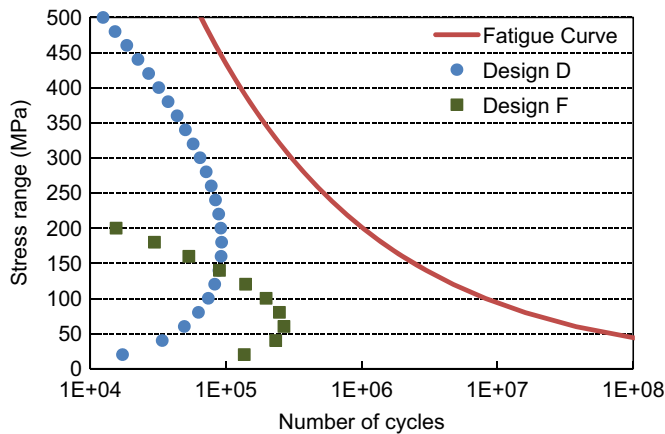


Fig. 17. Sample number of cycles for two designs (100 year life).

comfort level and the AASHTO deflection limit may have problems with the fatigue. The 100 years fatigue limit was almost the same as the England comfort level for this particular problem, and therefore, any solution that has smaller than 0.1 m displacement will meet all criteria. For illustrations, same number of cycles are shown in Fig. 17 for designs D and F. The fatigue curve is from the AASHTO Detail A. For any stress range, the number of cycles divided by the number given by the fatigue curve will provide fractional damage for that stress range. Following the Miner's rule, if cumulative damage of all stress ranges is greater than 1, the fatigue problem may arise. Damage index for Design D was 4.59 whereas the index for Design F was 0.19. The example provided here is to illustrate the importance of fatigue criteria in vortex-induced vibration. Since a simplified approach was employed, more elaborate method involving finite element analysis should be used for more accurate analysis.

7. Conclusion and future work

When a bridge is retrofitted to improve its aerodynamic performance, both the displacement and the cost have to be minimized. In order to find the solution under these conflicting objectives, we formulated the problem as a multi-objective optimization problem. The non-dominated sorting genetic algorithm (NSGA-II) was employed, along with the frequency-domain analysis of vortex-induced vibrations. The optimization produced the Pareto-optimal solutions, which were shown to be very useful in the decision-making. We also studied the solution considering the acceptance criteria for the allowable displacement, according to the North American Standard and the England Standard. The importance of checking the fatigue damage is also illustrated using a simplified fatigue analysis.

Further research is under way to improve the formulation of the multi-objective optimization. The current design space shown in Eq. (11) requires pre-defined number of groups of retrofit. A more general formulation is desirable. Also, the proposed approach can be extended to include other aerodynamic responses such as buffeting and flutter. Since the proposed framework is not specific to certain aerodynamic response, it will not be difficult to include any other response. However, when more than three objectives need to be used, further research in multi-objective optimization is necessary to obtain converged solutions in realistic time frame.

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